

R1 (10Points) Why is circuit switching used instead of packet switching in the public telephone network? Why is packet switching more efficient in the Internet infrastructure?

Because when telephone first built, it's transmitted by analog signal, with circuit switching, the network can guarantee the service in a good, continuous quality, and without long package delay. Packet switching on the other hand is implemented on our internet, and is more efficient because it won't waste any resource if the particular sender is not sending data. Other people could use the same frequency or same time-division slot to transmit at the same time.

R2 (10Points) List and describe the sources of delay in the Internet.

Propagation delay – the delay time which occurs when the package are transmitted through the hardware, such as the wire. It may occur some physical, collision delay. It's about the distance.

Transmission delay – The time which router need to push all the data outside, it's about the transmission rate and the packet's size.

Queuing delay – if the receive router can't handle all the data that fast, the arriving g packet has to wait at the buffer

Processing delay – the time which the router analyze the header, encode or decode the packet.

R3 (10Points) Describe the non persistent and persistent connection approach in HTTP. What are the pros and cons of the two approaches?

When Client ask a page downloading from server, the element is transferred through TCP. With non persistent mechanism, TCP has to establish a connection every time when the server needs to download each element, which will cause a lot of RTT. With persistent connection, TCP only request one connection, and transfer all the elements, which will save a lot of Acking(checking is the client ready for receiving packet)

What is the cons of persistent connection?

R4 (15Points) Briefly describe the performance requirements of real-time multimedia applications. Why is Forward Error Correction preferable to packet retransmission? Briefly describe the interleaving technique and explain why it is used. R5 (10Points)

Real-time multimedia such as VoIP have really low tolerance for delaying and loss-package, so the basic requirements is to keep the download rate, and use fixed played out delay or adaptive played out delay to remove jitter.

Forward Error Correction is to transmit a next chunk's data in a lower quality. If in the future the packet is drop, the system will get the lower quality data from previous transmission and played. This mechanism won't affect the bandwidth.

First, original stream is putting the packet in order it works is like – 1234 5678 9 10 11 12, instead of transmitting in order, the packet is divided separately. By doing so, when the packet is transmitted, even if some of it is lost, after it is reconstructed, there won't be a big part missed.

Concisely describe the client-server and the peer-to-peer application architectures and highlight their differences.

Client – server is an architecture of most website, server is running all the time, if the client request, server will response. In different type of service, HTTP, FTP. In P2P, every

individual node is server and client at the same time, it won't need a very high level infrastructure to handle a heavy weight of requesting, and it won't need to download the file in order.

R6 (10Points) With HTTP streaming, are the TCP receive buffer and the clients application buffer the same thing? If not, how do they interact?

Definition

No, they aren't the same thing.

On the client side, the client application (media player) reads bytes from the TCP receive buffer (through its client socket) and places the bytes into the client application buffer. At the same time, the client application periodically grabs video frames from the client application buffer, decompresses the frames, and displays them on the user's screen

R8 (20Points) List and describe Internet protocol stack.

Physical layer “ where bits moves, and all those hardware and wire's protocol.

Datalink layer “ Where switch or hub transmit ARP with mac address.

Network layer “ Where router forward the IP packet

Transport layer “ TCP, UDP send segments

Session layer “ connecting the conversation between two host

Presentation layer “ encoding the file type

Application layer - the application we use today, like browser